Student workbook

Introduction to Python and pandas

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This workbook goes with the Jupyter notebooks in the class repository. Type your answers in the notebook practice cells, or make notes here. The supplied orders data comes from Tech With Tim's PanadasTutorial.

Before the class

1. Clone https://github.com/Cyberpunk738/Data-Science-tutorial.git
2. Follow README.md to install requirements in .venv.
3. In desktop VS Code, install the Microsoft Python and Jupyter extensions.
4. Open notebooks/00_setup_check.ipynb and select the .venv kernel.
5. Confirm the ready message says 40 orders and 9 columns.

What we will do

Learn Python basics, read an orders CSV using pandas, select and filter records, count labels and save a result. Charts belong to a later class.

Key words

Word	Meaning
Python	The programming language
Interpreter	Software that runs Python code
Kernel	The running environment behind notebook cells
pandas	A library for working with tables
pd	The alias used for pandas
DataFrame	A labelled table
Series	A labelled one-dimensional collection
CSV	A text file with comma-separated fields

Part 1: Python foundations

P1. Your first output

Write code to display your name and one sentence about what you want to learn.

```
# Your answer:
```

Predict before running

```
quantity = 3
print(quantity)
print("quantity")
```

What does each line display? Why are the results different?

Notes: _____

_____

Types

Value	Your predicted type
3	_____
45.0	_____
"Alice Wong"	_____
True	_____

P2. Add two quantities

Store 3 and 5 in variables, then print their total.

```
# Your answer:
```

Lists and comparisons

For countries = ["USA", "Canada", "UAE"], what do countries[0], countries[1] and len(countries) return?

Notes: _____

For quantity = 20, predict quantity > 20, quantity >= 20 and quantity != 20.

Notes: _____

Part 1 continued

P3. A simple decision

Write an if/else statement: display Multiple units when quantity is at least 2; otherwise display One unit or fewer. Try quantity = 1.

```
# Your answer:
```

Explain the symbols

Symbol	Your explanation
=	_____
==	_____
!=	_____
>=	_____
[]	_____
()	_____

Before pandas

What is a library? What is the difference between installing pandas and importing it?

Notes: _____

_____

_____

Part 2: The orders table

Open notebooks/02_pandas_first_class.ipynb. Run the path helper and import/read cells. Keep the exact column spelling and capitalisation.

What does a row represent?

One row represents: _____

Question	Your answer
How many rows?	_____
How many columns?	_____
What does pd mean?	_____
What does df refer to?	_____
Why df.columns rather than df.columns()?	_____

Practice A. Several columns

Show Product, Category and Quantity for the first five orders.

```
# Your answer:
```

One row

Write code to get the first row by integer position. Which positions does df.iloc[0:3] select?

```
# Your answer:
```

Notes: _____

Filter quantities

Predict the difference between Quantity > 20 and Quantity >= 20. Run both and record the customers that appear.

Notes: _____

_____

Part 2 continued

Practice B. Orders from USA

Select only USA orders. How many rows match?

```
# Your answer:
```

Counts and meaning

What does len(df["Country"]) count? How does value_counts() differ? How would you count distinct country labels?

Notes: _____

_____

Practice C. Save Stationery orders

Select Stationery orders. Save them to OUTPUT_DIR / "stationery_orders.csv" without the pandas row index. Confirm the source CSV is unchanged.

```
# Your answer:
```

Why is False capitalised? What would index=True include?

Notes: _____

Exit questions

Explain one selection and one filter to a partner. How would you restore the original table after experimenting? What would you like to analyse next?

Notes: _____

_____

Part 3: Extra practice

Optional after the first class. Use notebook 03. Each update exercise starts with a copy.

Practice D. Combined and text filters

Show orders that are both Stationery and from Canada. Then show customer names beginning with J. Explain where each pair of parentheses belongs.

```
# Your answer:
```

Practice E. Update, rename and export

Starting from df.copy(), change every No in Shipped to Pending, rename CustomerName to Customer, and save outputs/shipping_practice.csv. Check df remains unchanged.

```
# Your answer:
```

Correct the preparation shorthand

Shorthand	Your corrected form
df.columns()	_____
df.iloc() for first row	_____
startwith	_____
rename(column=...)	_____
index=false	_____

Try before opening teacher/solutions.ipynb. The solutions are public for review after practice.

Dataset: https://github.com/techwithtim/PanadasTutorial

Class repository: https://github.com/Cyberpunk738/Data-Science-tutorial